APPENDIX I. TAJIKISTAN: DEBT SUSTAINABILITY ANALYSIS (DSA) – AN UPDATE

This update by Fund staff incorporates new developments since the DSA prepared at the time of the 2009 Article IV consultation and Tajikistan's request for access under the ECF (Country Report No. 09/174), and reflects the latest guidance on preparing DSAs. The update shows that when remittances are taken into account as foreign exchange inflows, the vulnerability indicators are substantially lower than previously assessed and remain below their respective thresholds. However, this does not change Tajikistan's rating of being at high risk of debt distress, which is based on indicators that do not account for remittances.

I. MAIN CHANGES SINCE THE PREVIOUS DSA

1. **The main changes regarding the macroeconomic outlook and assumptions since the previous DSA include:**

- Lower GDP growth to reflect the impact of the global crisis (by a half percentage point);
- External borrowing on concessional terms to finance $560 million in foreign exchange needs during 2011–14 for the Roghun HPP construction at IDA terms, which increases public debt by about 7 percent of GDP by 2014;
- Augmented access under the Fund's Extended Credit Facility (ECF) from 90 to 120 percent of quota, resulting in additional disbursements of SDR 26.1 million;[1] and
- Fiscal costs associated with recapitalization of the central bank in the amount SM 1.3 billion (interest and principal), or 2.7 percent of 2014 GDP.

2. **This update also incorporates remittances as a foreign exchange inflow comparable to exports, in line with the revised DSA guidelines.** In Tajikistan, remittances mainly from Russia, have proved to be a fairly stable and significant source of income, averaging 30–40 percent of GDP over the past years. Under these circumstances, remittances can be included in the denominators of debt burden indicators, following the modified DSA methodology.

[1] In addition, Tajikistan received equivalent of $128 million as part of the new SDR allocation to IMF members, but used only about $17 million in 2009 to retire expensive external debt.

II. External Debt—Risk Assessment

The revised DSA shows improved debt sustainability when including remittances in the calculations of the debt indicators, with only the extreme shock scenario breaching the respective thresholds. Still, Tajikistan remains at high risk of debt distress, given that in a DSA without remittances, there are prolonged breaches of the policy thresholds.[2]

3. **External debt ratios remain below the policy thresholds when remittances are included into the denominators of the modified debt burden ratios (Figure 1).** Compared to the original DSA (without remittances), the modified DSA (with remittances) shows that the debt ratios under the baseline scenario remain under the policy thresholds, and that breaches occur only under the most extreme shocks and under high investment-low growth scenario for debt-to-GDP plus remittances ratio.

4. **However, Tajikistan's risk of debt distress remains high because of significant and prolonged breaches of thresholds under the baseline scenarios and stress tests in the DSA without remittances (Figure 2).** In particular, the debt-to-export ratio significantly exceeds the threshold throughout the projected period, and the debt-to-GDP ratio breaches the threshold during 2014–19, albeit only marginally.

5. **Compared to the previous DSA, the debt indicators in the DSA without remittances have worsened, in part reflecting revised external financing needs for the Roghun HPP while making very conservative assumptions with respect to the project's impact on growth.** The worsening is mainly due to additional external borrowing to finance Roghun, as well as a lower discount rate applied in the DSA**.** Other effects come from downward revision to growth projections, and further domestic currency depreciation.[3] Alternative scenarios and stress tests demonstrate additional vulnerabilities of debt outlook to lower GDP growth, real exchange rate shocks, and nonconcessional or new concessional borrowing, and point to the need for the government to exercise extreme vigilance when incurring new debt, making sure that projects with the highest growth dividends are selected. Regarding Roghun, it should be noted that in the absence of a feasibility study, conservative estimates of Roghun's impact on growth have been used in this DSA. These estimates will likely be revised upward once the feasibility study, currently undertaken by the World Bank, is completed.

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[2] DSA with (without) remittances refers to DSA, in which remittances are incorporated (not incorporated) in denominators of debt burden indicators,

[3] Compared to the last DSA, the discount rate was revised downwards from 5 to 4 percent to reflect new global financial conditions.

6. **With respect to the proposed ECF augmentation from 90 to 120 percent of quota, the DSA suggests that no serious additional risks to debt sustainability arise.** First, the ECF is currently provided at zero interest rate.[4] But more generally, given that the authorities are committed to using the Fund disbursements to build reserves, the ECF leaves the country's net wealth position unchanged. Furthermore, in the absence of additional Fund financing, the BOP risks increase with potentially serious negative effects on growth.

III. PUBLIC DEBT

7. **The DSA demonstrates that public debt is sustainable, even after incorporating the costs of recapitalization of the central bank, however, risks could increase if fiscal deficits are not contained and/or growth dividends from high investments do not materialize (Figure 3).** Both the baseline scenario and the scenario incorporating additional debt issuance to implement the cotton debt resolution strategy show that public debt remains sustainable. Under the cotton debt resolution strategy, the government is expected to compensate commercial banks and the NBT for losses incurred as a result of financing of the cotton sector. As noted above, the costs associated with the recapitalization of the NBT are reflected in the baseline scenario, according to the estimates of these costs provided by the recent MCM mission. The public DSA's cotton debt resolution scenario reflects additional costs associated with issuance of bonds (SM 500 million) to banks in compensation for losses from past financing of the cotton sector at the refinancing rate of the NBT.[5] Public debt is unsustainable under low growth and/or fixed primary balance scenarios. The latter highlights the risk associated with insufficient fiscal consolidations compared to that assumed in the baseline scenario.

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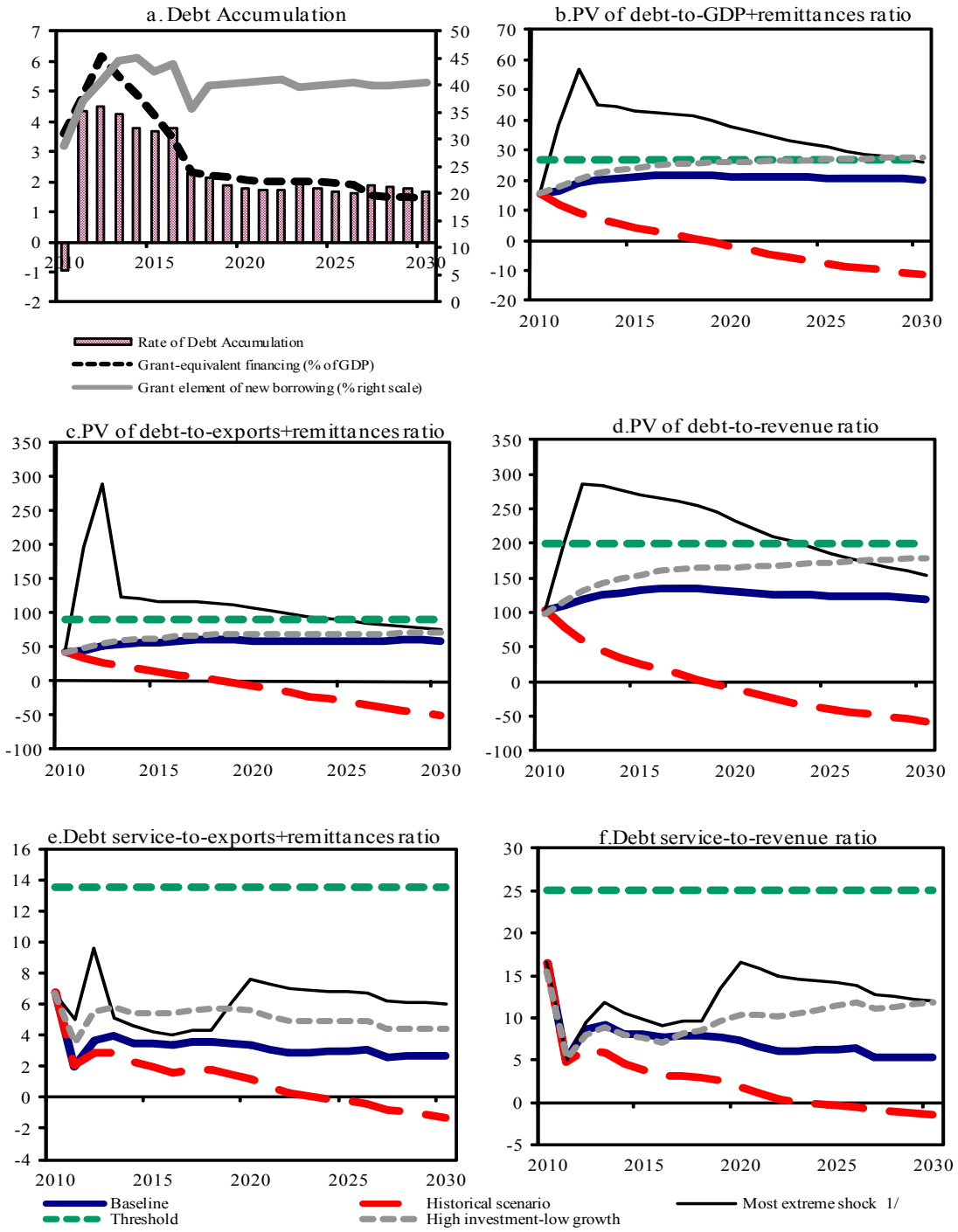
[4] The Fund reviews the level of interest rates for all concessional facilities every two years.

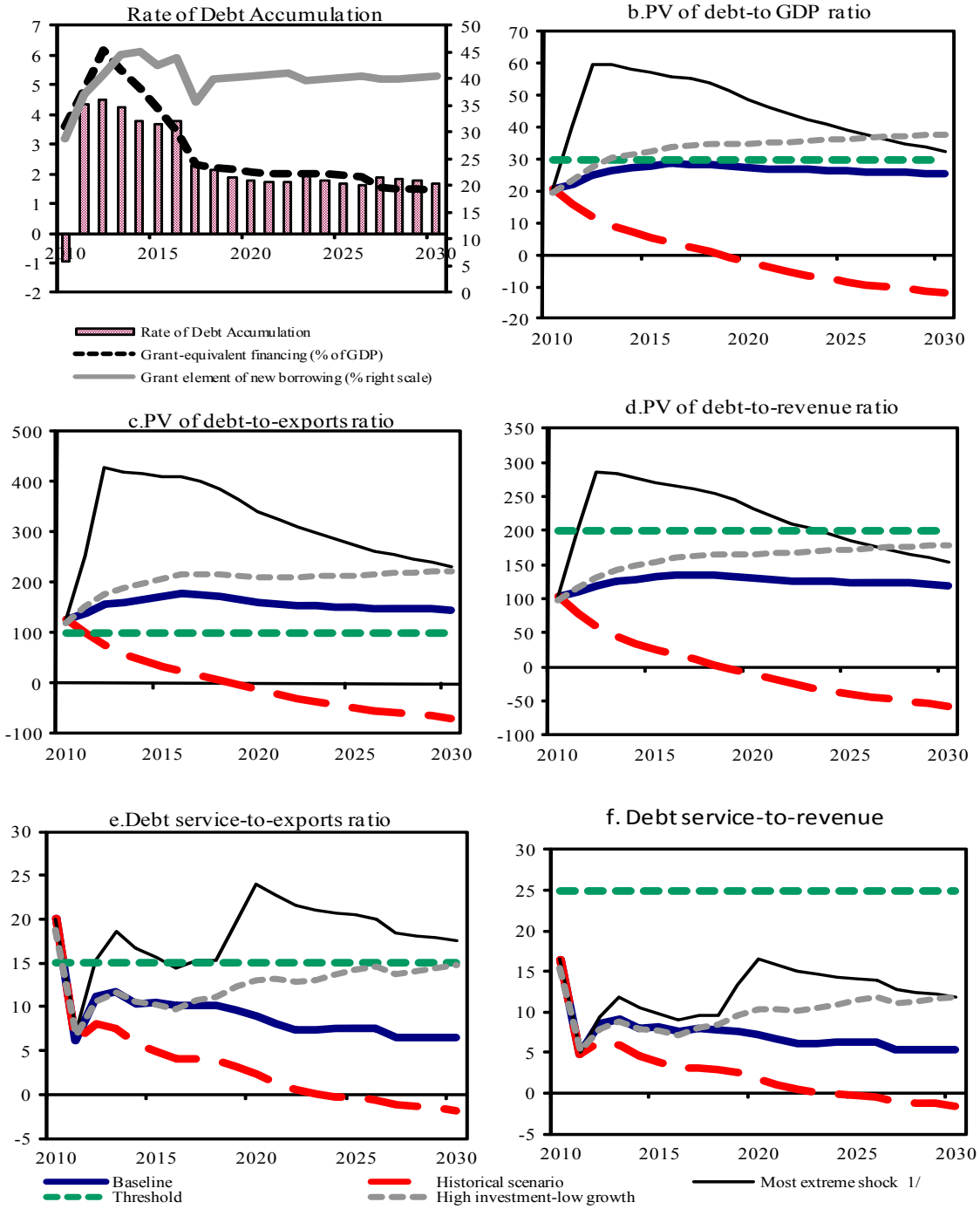
[5] The authorities currently intend to issue government bonds carrying 2 percent interest rate (i.e. much lower rate than assumed in the DSA).

Figure 1. Tajikistan: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios Incorporating Remittances, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a Non-debt flows shock; in e. to a Non-debt flows shock and in figure f. to a Non-debt flows shock.

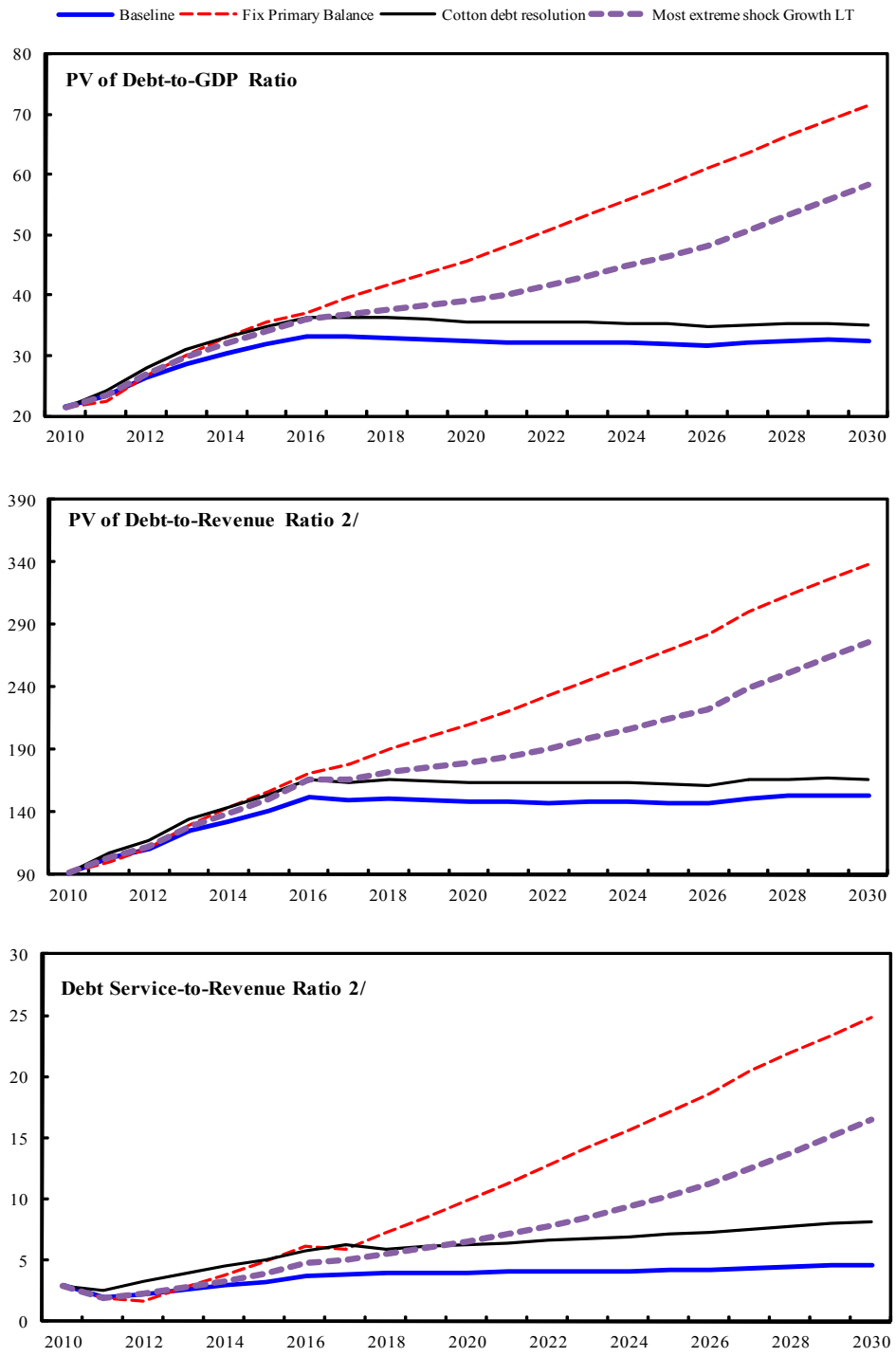
Figure 2. Tajikistan: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Non-debt flows shock; in c. to a Combination shock; in d. to a Non-debt flows shock; in e. to a Combination shock and in figure f. to a Non-debt flows shock.

Figure 3.Tajikistan: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Table 1. Tajikistan: External Debt Sustainability Framework, Baseline Scenario, 2007–29 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections								
	2007	2008	2009			2010	2011	2012	2013	2014	20010-2014 Average	2019	2029	2015-2029 Average
External debt (nominal) 1/	41.2	46.1	49.9			51.5	55.7	58.4	61.0	62.6		57.2	44.8	
o/w public and publicly guaranteed (PPG)	34.0	29.0	32.2			31.9	36.3	39.2	41.0	42.2		39.8	36.1	
Change in external debt	-2.8	5.0	3.8			1.6	4.2	2.7	2.7	1.6		-2.5	-1.3	
Identified net debt-creating flows	-6.3	-9.6	5.0			4.6	4.2	3.3	2.3	1.8		-0.8	0.8	
Non-interest current account deficit	6.8	6.4	5.9	1.5	3.6	6.8	7.1	7.7	7.9	8.0		4.9	3.8	4.9
Deficit in balance of goods and services	48.2	55.3	40.3			41.0	40.2	40.5	40.4	40.9		33.7	28.6	
Exports	20.7	16.8	16.4			15.9	15.8	16.3	16.1	16.1		16.8	17.2	
Imports	68.8	72.1	56.7			56.9	56.0	56.8	56.6	56.9		50.5	45.8	
Net current transfers (negative = inflow)	-41.9	-48.7	-34.4	-24.0	15.4	-34.3	-32.9	-32.5	-32.2	-32.5		-28.3	-24.2	-27.3
o/w official	-1.8	-3.1	-1.8			-1.7	-1.1	-0.9	-0.7	-0.6		-0.1	0.0	
Other current account flows (negative = net inflow)	0.5	-0.2	0.0			0.2	-0.2	-0.3	-0.3	-0.4		-0.6	-0.5	
Net FDI (negative = inflow)	-4.3	-5.8	-0.7	-3.7	3.6	-1.6	-1.8	-3.2	-4.3	-4.7		-3.9	-2.0	-3.3
Endogenous debt dynamics 2/	-8.8	-10.2	-0.3			-0.6	-1.1	-1.2	-1.3	-1.4		-1.8	-1.1	
Contribution from nominal interest rate	1.8	1.2	1.3			1.2	1.2	1.3	1.3	1.3		1.3	0.9	
Contribution from real GDP growth	-2.6	-2.4	-1.6			-1.8	-2.3	-2.5	-2.6	-2.7		-3.0	-1.9	
Contribution from price and exchange rate changes	-8.1	-9.1	…			…	…	…	…	…		…	…	
Residual (3-4) 3/	3.5	14.5	-1.2			-3.0	-0.1	-0.6	0.4	-0.2		-1.6	-2.1	
o/w exceptional financing	0.0	0.0	-0.3			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	39.5	40.9			44.2	46.2	47.0	48.4	49.3		44.8	33.7	
In percent of exports	...	234.7	249.6			277.6	292.0	289.0	299.8	306.4		266.9	195.5	
PV of PPG external debt	...	22.4	23.3			24.6	26.9	27.8	28.4	28.8		27.5	24.9	
In percent of exports	...	133.0	141.9			154.3	169.6	170.9	175.9	179.2		163.5	144.8	
In percent of government revenues	...	109.3	116.6			122.0	130.1	133.6	136.4	138.6		132.0	119.9	
Debt service-to-exports ratio (in percent)	13.2	15.2	24.4			10.9	16.1	16.5	15.2	15.4		13.2	8.5	
PPG debt service-to-exports ratio (in percent)	8.8	14.2	20.0			6.5	11.4	12.0	10.5	10.6		9.1	6.5	
PPG debt service-to-revenue ratio (in percent)	8.9	11.7	16.5			5.1	8.8	9.4	8.2	8.2		7.3	5.4	
Total gross financing need (Billions of U.S. dollars)	0.2	0.2	0.5			0.4	0.5	0.5	0.4	0.5		0.4	0.9	
Non-interest current account deficit that stabilizes debt ratio	9.6	1.5	2.2			5.2	2.9	5.0	5.2	6.4		7.4	5.2	
Key macroeconomic assumptions														
Real GDP growth (in percent)	7.8	7.9	3.4	8.1	2.1	4.0	5.0	5.0	5.0	5.0	4.8	5.5	4.5	4.8
GDP deflator in US dollar terms (change in percent)	22.5	28.2	-6.2	8.6	14.3	6.1	5.4	4.3	5.7	5.7	5.5	2.9	3.4	3.1
Effective interest rate (percent) 5/	5.5	4.2	2.8	4.4	1.2	2.6	2.7	2.6	2.5	2.4	2.6	2.3	2.0	2.2
Growth of exports of G&S (US dollar terms, in percent)	16.9	12.8	-5.5	4.3	22.5	7.2	10.0	12.5	10.2	10.7	10.1	9.7	7.9	8.5
Growth of imports of G&S (US dollar terms, in percent)	57.9	45.0	-23.7	17.1	25.8	10.8	8.9	10.9	10.6	11.7	10.6	6.8	6.7	6.5
Grant element of new public sector borrowing (in percent)	...	...	28.9	...	...	37.2	40.7	44.6	45.2	42.7	42.1	40.5	40.7	40.3
Government revenues (excluding grants, in percent of GDP)	20.5	20.5	20.0			20.2	20.6	20.8	20.8	20.8		20.8	20.8	20.8
Aid flows (in Billions of US dollars) 7/	0.1	0.1	0.2			0.3	0.5	0.5	0.5	0.5		0.4	0.7	
o/w Grants	0.1	0.1	0.2			0.1	0.2	0.1	0.1	0.1		0.1	0.0	
o/w Concessional loans	0.0	0.0	0.0			0.2	0.3	0.4	0.4	0.4		0.3	0.7	
Grant-equivalent financing (in percent of GDP) 8/	...	...	3.6			4.8	6.1	5.5	4.8	4.2		2.1	1.4	2.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	86.2			54.1	56.8	55.3	56.8	54.4		50.6	42.2	47.8
Memorandum items:														
Nominal GDP (Billions of US dollars)	3.7	5.1	5.0			5.5	6.1	6.7	7.4	8.2		12.5	26.1	
Nominal dollar GDP growth	32.0	38.3	-3.0			10.4	10.7	9.5	11.0	11.0	10.5	8.6	8.1	8.0
PV of PPG external debt (in Billions of US dollars)		1.1	1.1			1.3	1.6	1.8	2.0	2.3		3.4	6.4	
(PVt-PVt-1)/GDPt-1 (in percent)			-0.9			4.4	4.4	4.1	3.6	3.5	4.0	1.7	1.7	1.9
Gross remittances (Billions of US dollars)	1.5	2.3	1.6			1.8	1.9	2.1	2.3	2.6		3.5	6.3	
PV of PPG external debt (in percent of GDP + remittances)	...	15.4	17.6			18.5	20.4	21.1	21.6	21.9		21.4	20.1	
PV of PPG external debt (in percent of exports + remittances)	...	35.8	47.5			50.6	56.4	58.1	59.6	60.1		61.0	60.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	3.8	6.7			2.1	3.8	4.1	3.6	3.6		3.4	2.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2a. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009–29
(In percent)

	Act.	Projections						
	2009	2010	2011	2012	2013	2014	2019	2029
PV of debt-to GDP ratio								
Baseline	23	25	27	28	28	29	27	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2009-2029 1/	23	18	14	11	9	7	-2	-12
A2. New public sector loans on less favorable terms in 2009-2029 2	23	25	29	32	34	36	37	39
A3. Alternative Scenario : High investment-low growth	22	26	29	32	33	34	35	37
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	23	23	25	26	27	27	26	24
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	23	26	31	33	33	33	31	26
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	23	28	35	37	38	38	37	33
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	23	43	61	61	59	58	49	32
B5. Combination of B1-B4 using one-half standard deviation shocks	23	40	59	59	57	56	48	33
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	23	33	36	38	39	39	38	34
PV of debt-to-exports ratio								
Baseline	142	154	170	171	176	179	163	145
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2009-2029 1/	142	113	91	69	54	41	-11	-69
A2. New public sector loans on less favorable terms in 2009-2029 2	142	160	183	200	214	223	222	226
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	142	150	162	166	171	175	160	143
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	142	210	335	339	344	346	307	253
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	142	150	162	166	171	175	160	143
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	142	267	385	373	367	358	290	185
B5. Combination of B1-B4 using one-half standard deviation shocks	142	271	447	435	430	422	346	232
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	142	150	162	166	171	175	160	143
PV of debt-to-revenue ratio								
Baseline	117	122	130	134	136	139	132	120
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2009-2029 1/	117	89	70	54	42	32	-9	-57
A2. New public sector loans on less favorable terms in 2009-2029 2	117	126	140	156	166	173	179	187
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	117	117	120	126	129	131	126	115
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	117	129	152	157	158	159	147	124
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	117	139	169	177	181	184	176	161
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	117	211	296	292	284	277	234	153
B5. Combination of B1-B4 using one-half standard deviation shocks	117	197	284	282	276	270	232	159
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	117	166	173	181	185	188	181	165

Table 2a. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009–29 (continued)
(In percent)

	Act.	Projections						
	2009	2010	2011	2012	2013	2014	2019	2029
Debt service-to-exports ratio								
Baseline	20	6	11	12	11	11	9	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2009-2029 1/	20	6	9	9	7	6	3	-2
A2. New public sector loans on less favorable terms in 2009-2029 2	20	7	11	13	12	12	17	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	20	7	11	11	10	10	10	9
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	20	9	19	21	18	17	20	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	20	7	11	11	10	10	10	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	20	7	14	16	15	14	21	14
B5. Combination of B1-B4 using one-half standard deviation shocks	20	8	17	20	18	17	25	18
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	20	7	11	11	10	10	10	9
Debt service-to-revenue ratio								
Baseline	16	5	9	9	8	8	7	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2009-2029 1/	16	5	7	7	5	4	2	-2
A2. New public sector loans on less favorable terms in 2009-2029 2	16	5	9	10	9	9	13	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	16	5	8	9	8	7	8	7
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	16	5	9	10	8	8	10	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	16	6	11	12	11	10	11	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	16	5	10	13	11	11	17	12
B5. Combination of B1-B4 using one-half standard deviation shocks	16	6	11	13	12	11	17	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	16	7	12	12	11	10	12	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	40	40

Sources: Country authorities; and IMF staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitely assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2b.Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009-2029
(In percent)

	Act.	Projections							
	2009	2010	2011	2012	2013	2014	2018	2019	2029
PV of debt-to-GDP+remittances ratio									
Baseline	18	19	20	21	22	22	22	21	20
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2009-2029 1/	18	14	11	9	7	5	0	-2	-11
A2. New public sector loans on less favorable terms in 2009-2029 2	18	19	22	25	26	27	29	29	31
A3. Alternative Scenario : High investment-low growth	1	26	29	32	33	34	35	35	37
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	18	18	19	20	21	21	21	21	19
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	18	20	24	25	25	25	25	24	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	18	20	24	26	26	27	27	27	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	18	41	58	46	45	44	40	38	26
B5. Combination of B1-B4 using one-half standard deviation shocks	18	35	52	43	42	41	38	36	26
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	18	23	25	26	27	27	27	27	26
PV of debt-to-exports+remittances ratio									
Baseline	48	51	56	58	60	60	62	61	60
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2009-2029 1/	48	38	32	26	21	16	0	-5	-50
A2. New public sector loans on less favorable terms in 2009-2029 2	48	52	61	68	72	75	83	83	94
A3. Alternative Scenario : High investment-low growth	34	34	33	32	32	32	29	28	24
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	48	49	54	57	58	58	61	60	59
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	48	58	76	79	80	80	82	80	75
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	48	49	54	57	58	58	61	60	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	48	209	300	127	124	120	113	108	77
B5. Combination of B1-B4 using one-half standard deviation shocks	48	145	221	115	114	110	106	102	77
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	48	49	54	57	58	58	61	60	59
PV of debt-to-revenue ratio									
Baseline	117	122	130	134	136	139	135	132	120
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2009-2029 1/	117	89	70	54	42	32	-1	-9	-57
A2. New public sector loans on less favorable terms in 2009-2029 2	117	126	140	156	166	173	181	179	187
A3. Alternative Scenario : High investment-low growth	110	129	141	152	157	161	170	169	180
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	117	117	120	126	129	131	129	126	115
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	117	129	152	157	158	159	152	147	124
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	117	139	169	177	181	184	180	176	161
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	117	211	296	292	284	277	248	234	153
B5. Combination of B1-B4 using one-half standard deviation shocks	117	197	284	282	276	270	245	232	159
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	117	166	173	181	185	188	185	181	165

Table 2b.Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009-2029
(In percent)

	Act.	Projections							
	2009	2010	2011	2012	2013	2014	2018	2019	2029
Debt service-to-exports+remittances ratio									
Baseline	7	2	4	4	4	4	4	3	3
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2009-2029 1/	7	2	3	3	3	2	2	1	-1
A2. New public sector loans on less favorable terms in 2009-2029 2	7	2	4	4	4	4	6	6	6
A3. Alternative Scenario : High investment-low growth	0	0	0	0	0	0	0	0	0
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	7	2	4	4	3	3	4	4	4
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	7	2	4	5	4	4	5	5	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	7	2	4	4	3	3	4	4	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	7	5	11	6	5	5	6	8	6
B5. Combination of B1-B4 using one-half standard deviation shocks	7	4	8	5	5	4	6	7	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	7	2	4	4	3	3	4	4	4
Debt service-to-revenue ratio									
Baseline	16	5	9	9	8	8	8	7	5
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2009-2029 1/	16	5	7	7	5	4	3	2	-2
A2. New public sector loans on less favorable terms in 2009-2029 2	16	5	9	10	9	9	13	13	13
A3. Alternative Scenario : High investment-low growth	16	5	9	10	9	8	10	11	12
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	16	5	8	9	8	7	8	8	7
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	16	5	9	10	8	8	9	10	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	16	6	11	12	11	10	11	11	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	16	5	10	13	11	11	14	17	12
B5. Combination of B1-B4 using one-half standard deviation shocks	16	6	11	13	12	11	14	17	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	16	7	12	12	11	10	11	12	11
Memorandum item:									
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	40	40	40

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Tajikistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007–29
(In percent of GDP, unless otherwise indicated)

	Actual					Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2010-14 Average	2019	2029	2015-29 Average
Public sector debt 1/	35.2	29.9	33.0			33.7	38.8	43.1	46.1	48.1		46.6	44.2	
o/w foreign-currency denominated	34.0	29.0	32.2			31.9	36.3	39.2	41.0	42.2		39.8	36.1	
Change in public sector debt	-0.7	-5.3	3.1			0.7	5.2	4.3	3.0	2.0		-1.0	-0.4	
Identified debt-creating flows	-3.6	-4.5	7.6			1.9	1.4	0.0	-0.5	-0.3		-0.7	0.1	
Primary deficit	5.0	4.4	4.7	1.2	2.8	5.4	3.6	3.7	3.3	3.7	3.9	2.4	2.8	2.7
Revenue and grants	22.5	22.1	23.3			22.6	23.5	22.7	22.6	22.4		21.5	20.9	
of which: grants	2.0	1.6	3.4			2.4	2.9	1.9	1.8	1.6		0.7	0.1	
Primary (noninterest) expenditure	27.5	26.5	28.1			28.0	27.1	26.4	25.9	26.0		23.9	23.7	
Automatic debt dynamics	-8.5	-8.6	2.9			-3.5	-2.2	-3.6	-3.7	-4.0		-3.0	-2.7	
Contribution from interest rate/growth differential	-2.3	-2.1	-1.0			-1.4	-1.8	-2.1	-2.4	-2.6		-2.7	-2.1	
of which: contribution from average real interest rate	0.3	0.5	0.0			-0.2	-0.2	-0.3	-0.4	-0.4		-0.2	-0.1	
of which: contribution from real GDP growth	-2.6	-2.6	-1.0			-1.3	-1.6	-1.8	-2.1	-2.2		-2.5	-1.9	
Contribution from real exchange rate depreciation	-6.2	-6.6	3.9			-2.1	-0.3	-1.5	-1.3	-1.4		...	...	
Other identified debt-creating flows	-0.1	-0.3	-0.1			-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1	-0.3	-0.1			-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.9	-0.8	-4.5			-1.2	3.8	4.2	3.4	2.4		-0.3	-0.5	
Other Sustainability Indicators														
PV of public sector debt	1.2	23.3	24.0			26.4	29.4	31.7	33.4	34.8		34.3	33.1	
o/w foreign-currency denominated	0.0	22.4	23.3			24.6	26.9	27.8	28.4	28.8		27.5	24.9	
o/w external	...	22.4	23.3			24.6	26.9	27.8	28.4	28.8		27.5	24.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...		...	...	
Gross financing need 2/	6.1	5.5	5.4			5.9	4.2	4.3	4.0	4.5		3.3	3.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	5.4	105.2	102.9			117.1	124.9	139.7	147.9	155.5		159.4	158.3	
PV of public sector debt-to-revenue ratio (in percent)	5.9	113.5	120.3			131.0	142.6	152.5	160.7	167.1		164.7	158.9	
o/w external 3/	…	109.3	116.6			122.0	130.1	133.6	136.4	138.6		132.0	119.9	
Debt service-to-revenue and grants ratio (in percent) 4/	4.8	4.9	2.9			2.2	2.4	2.8	3.2	3.4		4.1	4.6	
Debt service-to-revenue ratio (in percent) 4/	5.2	5.3	3.4			2.5	2.8	3.0	3.4	3.7		4.3	4.6	
Primary deficit that stabilizes the debt-to-GDP ratio	5.7	9.7	1.6			4.7	-1.6	-0.6	0.3	1.7		3.4	3.2	
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	7.8	7.9	3.4	8.1	2.1	4.0	5.0	5.0	5.0	5.0	4.8	5.5	4.5	4.8
Average nominal interest rate on forex debt (in percent)	3.5	4.4	2.1	3.2	0.7	1.6	1.7	1.7	1.6	1.5	1.6	1.5	1.5	1.5
Average real interest rate on domestic debt (in percent)	-9.2	-20.9	-3.8	-10.3	7.4	-4.7	-6.8	-6.2	-5.2	-5.2	-5.6	-0.5	0.3	-0.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-19.3	-20.4	14.0	-4.9	15.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	27.9	27.7	13.3	21.7	6.8	15.0	13.5	12.0	11.0	11.0	12.5	6.0	6.0	6.3
Growth of real primary spending (deflated by GDP deflator, in percent)	0.4	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	28.9	…	…	37.2	40.7	44.6	45.2	42.7	42.1	40.5	40.7	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Tajikistan: Sensitivity Analisys for Key Indicators of Public Debt, 2009–29

	Act.	Projections						
	2009	2010	2011	2012	2013	2014	2019	2029
PV of Debt-to-GDP Ratio								
Baseline	24	26	29	32	33	35	34	33
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	21	21	20	18	17	7	-6
A2. primary balance is unchanged from 2009	24	26	30	33	36	39	49	74
A3. Permanently lower GDP growth 1/	24	27	30	33	35	37	41	59
A4. Alternative scenario: Cotton debt resolution	24	28	31	34	36	37	36	36
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010-2011	24	25	28	29	30	31	27	18
B2. primary balance is at historical average minus one standard deviations in 2010-2011	24	25	28	30	32	33	32	31
B3. Combination of B1-B2 using one half standard deviation shocks	24	23	24	25	25	26	20	5
B4. One-time 30 percent real depreciation in 2010	24	36	38	38	39	40	39	39
PV of Debt-to-Revenue Ratio 2/								
Baseline	103	117	125	140	148	155	159	158
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	103	94	89	87	82	75	33	-28
A2. primary balance is unchanged from 2009	103	114	127	146	161	174	226	354
A3. Permanently lower GDP growth 1/	103	118	127	144	154	165	191	281
A4. Alternative scenario: Cotton debt resolution	103	126	134	149	157	165	170	171
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010-2011	103	113	117	129	134	138	126	85
B2. primary balance is at historical average minus one standard deviations in 2010-2011	103	109	117	131	139	147	150	146
B3. Combination of B1-B2 using one half standard deviation shocks	103	100	102	110	113	115	92	22
B4. One-time 30 percent real depreciation in 2010	103	162	161	169	172	177	180	188
Debt Service-to-Revenue Ratio 2/								
Baseline	3	2	2	3	3	3	4	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	3	2	-1	-2	-3	-5	-6	-10
A2. primary balance is unchanged from 2009	3	2	2	3	4	6	11	26
A3. Permanently lower GDP growth 1/	3	2	2	3	4	4	7	16
A4. Alternative scenario: Cotton debt resolution	3	2	4	4	5	5	5	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010-2011	3	2	2	2	2	2	1	-3
B2. primary balance is at historical average minus one standard deviations in 2010-2011	3	2	1	1	2	2	3	3
B3. Combination of B1-B2 using one half standard deviation shocks	3	2	0	0	-1	-1	-2	-10
B4. One-time 30 percent real depreciation in 2010	3	3	3	4	5	5	6	9

Sources: Country authorities; and IMF staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.